
National University of Computer & Emerging Sciences



AL2002 – Artificial Intelligence – Lab (Fall 2026) BSSE-5B

Lab Work 2

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Instructions:

Note: Carefully read the following instructions

1. You also have to submit .ipynb file and word file.
2. Plagiarism is strictly prohibited, 0 marks would be given to students who cheat.
3. Write codes in PYTHON language and you may use any IDE or Notebook environment.
4. First think about the problem statements before starting your programming.
5. At the end when you have done your tasks, attached .py or .ipynb and word files on google classroom.
6. Paste your complete code in word file along with output in case of .py file (Make sure your submission is completed). In case of missing any file, marks will be deducted.
7. Please submit your file in this format 23Fxxxx_Name_SecA_Lab#
8. Do not submit your assignment after deadline. Late and email submission is not accepted.
9. Do not copy code from any source otherwise you will be penalized with ZERO Marks.
10. **YOUR MARKING WILL BE BASED ON THE PRIOR SUBMISSION OF YOUR CODE BEFORE DEADLINE.**

1. Write a program in Python to swap two numbers. Take two variables “num1” and “num2”. Assign values to both variables. Swap both numbers without using a third variable. Display output using String formatting %d (use for integers).
2. Write a program to find if the number is the factor of 24 or not. Take input value of a number from the user and then check if it is factor of 24 then print “Input number is factor of 24” otherwise print “Input number is not a factor of 24”.
3. Take input of all the Departments of FAST CFD, then print last department assigned in the list. Display output using String formatting %s (use for string).
4. Write a python code to do the following task
 - i. Take 2 strings with sequence = “HELLO, WORLD” and “HOW, arE yOu”.
 - ii. Do the following tasks Concatenate the 2 strings, Make the 2 strings in upper and lowercase, Slice the strings using python string slicing. The following string should look like this after slicing; “, World” and “E yO”.
 - iii. For the second string you have to use only negative number for slicing. Note: Read the instructions carefully. And the output of string should be exact especially in slicing.
5. Take a start value and end value from user and print Even and Odd numbers between them using for and while loop.

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6. `list = ["apple", "cherry", "orange", "kiwi", "melon", "mango"]`
 - i. Remove "cherry" and "melon" from list.
 - ii. Add "banana" at second last index.
 7. `marks = {'AI':74,'CN':76,'DS':42,'PS':54}`
 - i. Print the sum of marks using for loop.
 - ii. Print keys and values both using for loop.
 - iii. Print max marks and relative subject name.
 - iv. Print mix marks and relative subject name.
 8. A **warehouse storage system** named "**StorageMax**" is used to efficiently allocate space for storing rectangular cargo containers. To optimize storage, the system must calculate the floor area occupied and the total storage volume available.

Create a class named "**StorageMax**", where:

- The user provides the **length, width, and height** of the storage unit.
- A member function **Calculate_area()** computes the **floor area** occupied by the unit.
- A member function **Calculate_volume()** computes the **total storage volume** available.
- A member function **display()** shows the calculated **area and volume**.
- At the end, the program prints: "**Storage Allocation Complete...**" **Note:** Use a **tuple** for input values.

9. A **fitness tracking system** named "**FitTrack**" categorizes workout routines into different types to help users manage their fitness plans efficiently. Perform various operations using built-in set functions.

Create a class named "**FitTrack**", which includes:

- Four sets representing different **workout categories: Cardio, Strength, Flexibility, Endurance**.
- A member function to **add an exercise** to a specific category.
- A member function to **remove an exercise** from a category.
- A member function to **check if an exercise exists** in a particular category.
- A member function to perform **set operations**:
 - **Union** (exercises included across multiple categories).
 - **Intersection** (exercises common in selected categories).
 - **Difference** (exercises that belong to one category but not another).

Note: Use **built-in functions** for sets in this task.

10. A **home safety assessment system** named "**SafeHome**" helps users determine if their home has potential safety hazards. The system evaluates different risk factors and determines whether action is needed. A dictionary stores predefined hazard severity scores.

Potential Home Safety Hazards:

- **Exposed electrical wiring:** 5
 - **Gas leak smell:** 5
 - **Water leakage or mold growth:** 4
 - **Loose stair railings:** 3
 - **Faulty smoke detectors:** 4
 - **Clogged or blocked exits:** 3
 - **Slippery floors without mats:** 3
 - **Windows or doors that don't lock properly:** 4
- Instructions:**
- Ask the user if they have noticed each hazard (**Yes = 1, No = 0**).
 - Sum the severity scores based on the user's input.
 - Define a **threshold of 15** to determine if urgent action is required.

Output Statements:

- If the total score is above **15**:
"**Based on the hazards reported, your home may require safety improvements. It is recommended to address these issues to prevent accidents.**"
- If the total score is below **15**:



**"Based on the hazards reported, your home does not seem to have critical safety risks.
However, regular maintenance is always a good idea."**